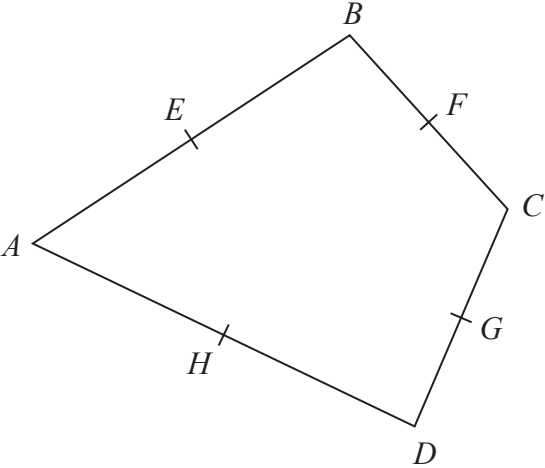




$ABCD$ is a quadrilateral.

E, F, G and H are the midpoints of AB, BC, CD and DA .

$$\overrightarrow{AH} = \mathbf{a} \quad \overrightarrow{AE} = \mathbf{b} \quad \overrightarrow{DG} = \mathbf{c}$$

Prove, using vectors, that $EFGH$ is a parallelogram.